Behaviour of Sandwich composites under flexural load – Effect of variation of core thickness

A PROJECT REPORT

Submitted by

SRIVYAS NARAYAN (21103008)

R MUTHIAH (21103028)

K MAGESH (21103010)

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

AEROSPACE ENGINEERING

DEPARTMENT OF AEROSPACE ENGINEERING

HINDUSTAN INSTITUTE OF TECHNOLOGY AND SCIENCE

PADUR, CHENNAI - 603 103

APRIL 2025

HINDUSTAN INSTITUTE OF TECHNOLOGY AND SCIENCE: PADUR, CHENNAI - 603 103

BONAFIDE CERTIFICATE

Certified that this project report titled **"Behaviour of Sandwich composites under flexural load – Effect of variation of core thickness"** is the Bonafide work of "**MUTHIAH R (21103028) SRIVYAS NARAYAN (21103008) K MAGESH (21103010)"** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported here does not form part of any other project / research work on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

HEAD OF THE DEPARTMENT	**SUPERVISOR**
Dr P VASANTHAKUMAR	**Dr R SANTHANAKRISHNAN**
PROFESSOR	**PROFESSOR**
School of Aeronautical Sciences	School of Aeronautical Sciences
Hindustan Institute of Technology & Science Science Padur, Chennai.	Hindustan Institute of Technology & Science Padur, Chennai.

The Project Viva-Voce Examination is held on ___________________

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

ACKNOWLEDGEMENTS

We would like to place on record our sincere thanks to all those who contributed to the successful completion of our final year project work. It is a matter of pride and privilege for us to express our deep gratitude to the management of Hindustan Institute of Technology and Science for providing us with the necessary facilities and support.

We express our deep sense of gratitude to our respected Chairperson Dr. **(Mrs.) Elizabeth Verghese** and Pro-Chancellor **Dr.Anand Jacob Verghese** for giving us an opportunity to do the project. We would like to thank our Director **Mr. Ashok Verghese** and Vice Chancellor **Dr. S.N. Sridhara** for giving us moral support to complete this project. We would like to express our grateful thanks to Dean (E&T) **Dr. Angelina Geetha** for support and encouragement.

We extend our sincere thanks to our Head of the Department **Dr. P. Vasanthakumar** for inspiring and motivating us to complete this project. We would like to thank our internal guide **Dr. R. Santhanakrishnan** for continually guiding and actively participating in our project, giving valuable suggestion to complete the project. We would like to thank all the faculty members of the School of Aeronautical Sciences, who have directly or indirectly extended their support. Finally, we deeply indebted to our parents who have been our greatest support while we worked day and night for the project to make it a success.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF SYMBOLS AND ABBREVATIONS

A	-	Average Cross Sectional Area (mm^2)
M	-	Bending Moment (N-mm)
σ_B	-	Bending Strength (Mpa)
c	-	Core Thickness
d	-	Depth of Specimen(mm)
d_c	-	Distance between facing centroids
W	-	Energy Absorption(J)
D_E	-	Experimental flexural stiffness (N-mm_2)
f	-	Face sheet thickness(mm)
m	-	Mass of the Sandwich panel (Kg)
P	-	Maximum load(N)
σ_f	-	Maximum normal stress (Mpa)
E_c	-	Modulus of core (Mpa)
E_f	-	Modulus of facing (Mpa)
δ	-	Relative displacement (mm)
L_s	-	Span Length(mm)
L	-	Specimen Length(mm)
T	-	Thickness of the specimen(mm)
B	-	Width of the specimen(mm)
FRC	-	Fibre Reinforced Composite
FRP	-	Fibre Reinforced Plastic
GFRP	-	Glass Fibre Reinforced Polymer
PUF	-	Poly Urethane Foam

ABSTRACT

The application of sandwich structures is widening day by day. Many critical areas such as aerospace, construction, marine, etc. are some of the important applications of sandwich structures. The use of composite structures in aerospace and civil structural applications have been increasing especially due to their extremely low weight that leads to reduction in the total weight, high flexural and corrosion resistance in addition to higher transverse shear stiffness. Various combinations of core and face sheet thickness have been evaluated by many researchers worldwide.

In this project we performed experimental investigation of flexural strength of Sandwich panels of varying core thicknesses . The Sandwich panels consist of face sheet and PVC foam cores of varying thickness and altogether are bound with Resin and Hardener mixture. The Sandwich panels were fabricated using Vacuum infusion. Flexural tests were then conducted according to ASTM standards to determine the variation of flexural strength with varying core thickness. Three point bending test to determine flexural strength was in accordance with ASTM standard C393.

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

In recent years, the research about the composite materials is continuously improving among the researchers, which can be used in various fields. A composite material is defined as a combination of two or more materials having different physical and chemical properties. Which when combined becomes stronger, lighter and improve strength and stiffness.

Composite materials are defined as the combination of two or more than two materials at a macroscopic level, it is done to develop some features that are not available or limited in conventional materials. Composite materials where the fibers are reinforced by polymer matrix are used in many industrial building materials such as aerospace, automobile, defence, and construction. Glass, carbon, aramid and boron is some of the synthetic fibers which are most used reinforcing fibers types. Because of the high strength to weight ratio and stiffness, composite materials find wide applications in aircraft structures, wind turbines, and marine vehicles. The structures used for marine applications experience fatigue loads due to wave action, aircraft structures due to the Ground to Ground (GAG) cycle. The structures used for marine and aerospace applications need to have good bending characteristics, bending stress, and mechanical strength. The composite materials used for structural applications are mostly thermosetting types in which the fibers such as carbon, glass, Kevlar are embedded into the epoxy resin.

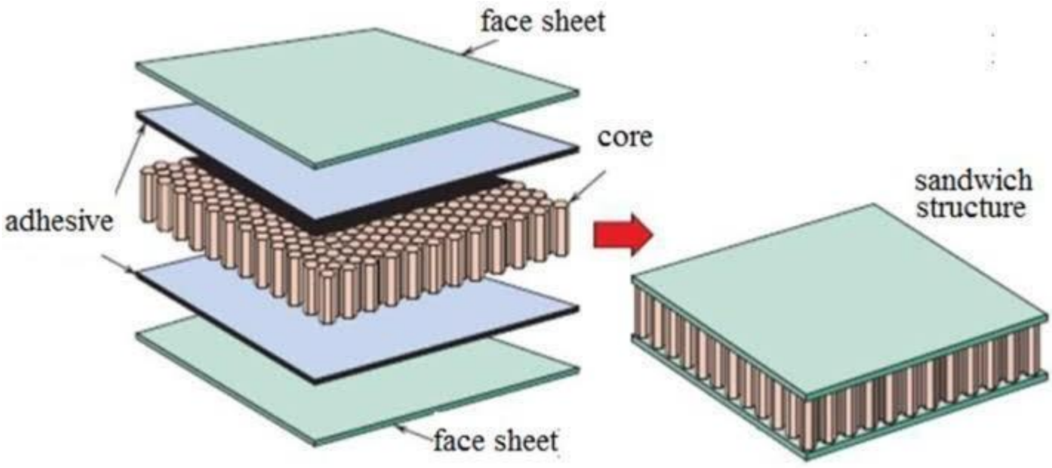


Figure 1.1 Composite Sandwich Structure

Sandwich panel using composites materials have been attracted by researchers due to high strength-to-weight ratio. A sandwich composite is when two stiff composite panels which are generally stiff and light in weight have been attached to a lightweight but thick core. The main reason behind the fabrication of composite sandwiches panel is to provide a composite with high bending stiffness with overall low density.

Sandwich panel consists of namely two face sheets and a core bound together by a resin and hardener mixture. The face sheet / skin is adhesively stuck to the core to gain a load transfer among the components. By this method the properties of each detached component have been combinly utilized to the structural advantages of the whole assembly which helps to achieve a very high stiffness to weight ratio. Typically, the facing layer will be laminates, aluminium plates, and glass fiber reinforced polymers. Basically, the skins considered are thin, very strong and stiff.

The sandwich structure core is the structure that is placed in between two thin faces. The material of the core of the sandwich structure is usually low strength material but also has higher thickness provides the sandwich composite with high bending stiffness and high strength to weight ratio and low density. The purpose of the core is to back the thin skin so that it does not get deformed and stays fixed in relative to each other. As the major need for the aviation industry is a material with high strength to weight ratio, sandwich panels with PVC foam core have good strength, light weight, high mass specific properties, high strength to weight ratio, etc. Although there are works on the mechanical properties of many sandwich panels, the amount of work that has been focused on the varying core thickness of sandwich panels and their corresponding variation of flexural strength hasn't been performed for PVC foam core. Thus, our work deals with studying the variation of flexural strength with variation of core thickness for glass fibre-based sandwich panel.

From the literature survey, it can be noted that there are many works related to the mechanical properties of sandwich panel, whereas only a few works were related to flexural strength variation of sandwich panels. So, our work was to evaluate the behaviour of Sandwich composites under flexural load by varying the core thickness in the present research work, the sandwich panel was made using vacuum infusion method with PVC foam as core and glass fibre as face sheet. The panel was tested for their bending strength to calculate flexural rigidity.

1.1 OBJECTIVE OF THE WORK

The objective of the present study is to evaluate the behaviour of Sandwich composites under flexural load by varying the core thickness.

1.3 SCOPE OF THE WORK

The broad scope of the project as envisaged are

- To conduct the experimental and numerical investigation of the bending strength of sandwich panel with varying core thickness.
- To compare the experimental results of variation of flexural strength of varying core sandwich panel.

1.4 THESIS ORGANISATION

This project is to evaluate the behaviour of Sandwich composites under flexural load by varying the core thickness. The thesis presentation is organized as shown below.

Chapter 1: This chapter deals with the background studies carried out with the sandwich structure and corrugated core objective, problem statement and scope.

Chapter 2: This chapter deals with the summary of review of literature.

Chapter 3: This chapter deals with the materials and methods used in this project.

Chapter 4: This chapter discusses the testing methods for flexural strength.

Chapter 5: This chapter discusses about experimental results.

Chapter 6: This chapter gives a summary of the work and major conclusions from the study.

1.5 SUMMARY

In this chapter, a brief description of the sandwich structures, motivation, objective, problem statement, and scope are discussed in detail.

CHAPTER 2
LITERATURE SURVEY

2.1 INTRODUCTION:

As composite materials find wide use in aircraft structures as main loading carrying members, the need to understand the material characteristics increased over the year. A considerable amount of research is being carried out in this domain in recent years. A literature survey has been done for sandwich structure construction and bending tests performed on corrugated core sandwich panels of different shapes.

2.2 EARLY WORK:

L. J. GIBSON et al(1989) made sandwich panel having carbon fibre face sheets with Titanium and carbon fibre pins being inserted into the polyetherimide foam core to increase the through-thickness strength. It was found that the foam core helps in stabilising the pins against elastic buckling, and the strength and energy absorption capacity of the pin reinforced core has been more than their individual contributions from the foam and unsupported pins and concluded that the micro-inertia of the pins stabilises the panel against elastic buckling and leads to the observed elevation in strength.

T. C. Triantafillou et al.,(1989) The study analyzes debonding in foam-core sandwich panels using strain energy release rate criteria. Experiments with aluminum faces and polyurethane foam cores show debonding occurs only with significant pre-existing cracks. The findings help determine permissible interface crack sizes and compare debonding with other failure modes like face yielding and core shearing.

M Akaya et al.,(1990) The paper compares honeycomb-core and foam-core carbon-fiber/epoxy sandwich panels for aircraft flap construction. It evaluates manufacturing techniques, low-energy impact damage, residual compressive strength, and impact puncture resistance. Honeycomb provides structural efficiency but risks buckling, while Rohacell foam offers design flexibility but may collapse under pressure. Higher-density foam improves impact resistance.

William S et al., (1996) presented a model for the dynamic compressive layer of polymer matrix fiber composites. Here the model is examined for suddenly applied constant compressive load. Under constant load it's found that inertial effects contribute to a reduction in the critical stress for composite failure. The reduction has been greatest for composites with long initial fiber imperfection wavelengths. For any given load, there will be a range of initial fiber imperfection wavelengths which would result in composite failure.

SJ Amith Kumar et al., (2004) conclude that Truss core construction appears to be as efficient as honeycomb core construction for sandwich plates. Considering the competition between the two methods of construction in terms of weight perspective, the advantage outcome is likely to alter on other issues such as manufacturing process, vulnerability to delamination or moisture, and other multifunctional capabilities.

Y.S. Tian et al., (2005) considered a total of eight different panel geometries including hat and blade-stiffened panels, and square, triangular and trapezoidal cores. From a weight standpoint, it has been found that panels with hat-stiffeners are found to be the most efficient for the given boundary conditions and sandwich panels with a square core are found to be the least efficient.

A. Vaziri et al., (2006) studied square honeycomb and folded plate steel cores filled with two densities of structural foams and found that the plates with foam filled cores performs nearly same as plates of the same weight with unfilled cores. So, the decision on use of foams in the cores depends upon multifunctional advantages such as acoustic and thermal insulation or environmental isolation of core interstices.

Chengfu Shua et al., (2006) investigated the multi-layered sandwich panels under the quasi-static crushing loading. It was found that the sandwich configuration and number of layers considered for the core played an important role in analysing the failure mechanism and energy absorption. It was concluded that the cross-arranged panels are better than the regular-arrange and stagger arranged panels not only in the energy absorption but also in the aspect of peak crushing force.

Styles et al (2007) studied the effect of core thickness on the deformation mechanism of aluminum foam core composite facing sandwich panel using four point bending test. It was found from the results thicker samples are failed by core indentation and thinner samples are failed by skin wrinkling and fracture, and some crushing and core cracking.

Engin et al (2008) investigated the material characteristics of 3-D FRP sandwich panels. The fundamental characteristics are compression, tension, flexure and shear are studied under the influence of the panel thickness, through-thickness fiber configuration and density. The stiffness reduction was approximately 33% and due to increasing density through thickness fiber waves among the fibers was generated which in turn reduces the strength of the face sheets. The tensile strength is decreased is also significant because of increasing the amount of thickness fiber insertion. This study also revealed that the thickness of the fiber does not affect the initial core shear modulus but reduces the shear strength.

Dan Zenkert et al., (2009) performed fatigue test on sandwich beams. It was found that for high load and small number of cycles to failure, the beams fail by face tensile fracture. For lower loads, and large number of cycles to failure, the beams fail by core shear.

Benjamin P. Russel et al., (2010) considered two variants of the sandwich cores. The sandwich core with the empty spaces between the corrugations filled with a PVC foam, and unfilled corrugations were investigated. The compression tests on both the corrugated cores and the parent strut wall material confirmed that these relatively high relative density corrugated cores failed by micro buckling of the strut wall material under quasistatic loading. The foam which has been used for filling did not have any significant effect on the measured responses. The peak stresses of both the strut wall material and corrugated cores increased linearly with strain rate less than 4000 per second.

V Rubino et al., (2010) manufactured sandwich beams comprising y-frame and corrugated cores in a way that the longitudinal axis of the cores coincides with the axis of the beams. The quasi-static three-point bending response was measured along with the indentation response of the beams placed on a rigid foundation. Here it has been observed that the simply supported beams are displaying a softening response beyond the initial peak load whereas the clamped beams display a hardening response because of the longitudinal stretching of the face-sheets.

Iman Dayyani et al., (2011) compared tensile and flexural characteristics of laminate panels with numerical and analytical methods. The inference is that the mechanical behaviour of the core in tension is sensitive to the variation of core height. Also, it has been found that the tensile strength of corrugated core is 16.61 times more than the bending strength of the core.

Mitra et al (2012) investigated an innovative method to improve the delamination resistance capacity of initially delaminated sandwich composites subjected to in-plane compressive loading by inserting the premanufactured shear keys in foam core grooves.

Ali Shabafan et al.,(2013) The study investigates how processing parameters (press temperature, pressing, and foaming times) affect the physical and structural properties of lightweight foam-core sandwich panels. Results show that optimizing these parameters improves dimensional stability, surface quality, and water resistance. Proper adjustments enhance foam-cell structure, reducing swelling and absorption while maintaining lightweight efficiency.

Jian Xiong et al., (2013) studied Bending properties and failure modes of carbon fiber composite egg and pyramidal honeycomb beams by doing three-point bending tests. The finite element method was employed to determine the ratio of maximum displacement to applied load of sandwich beam with two different honeycomb cores. The bending test results has been very useful for designing sandwich beams with lightweight structure and multifunctional applications.

Jin Zhang et al., (2013) investigated the bending strength, stiffness and energy absorption of corrugated sandwich composite structure to explore novel designs of lightweight load-bearing structures that are capable of energy absorption in transportation vehicles. The results revealed that the hybridization of glass fibres and carbon fibres taken in the ratio of 50:50 in face-sheets was able to compete the equivalent specific bending strength as the face-sheets made entirely of carbon fibre composites. The composite core with hybrid structure and with foam insertion showed medium energy absorption, ranging between the energy absorption values of glass fibre and the carbon fibre composite coupons, and also displayed the highest crush force efficiency among all designs.

Okan et al., (2014) The study examines the impact response of sandwich composites with poly(vinyl chloride) (PVC) and poly(ethylene terephthalate) (PET) foam cores of varying thicknesses. Experimental results show that increasing core thickness improves energy absorption and impact resistance. PET and PVC foams exhibit different failure mechanisms, affecting penetration and perforation thresholds in composite structures.

M. Nawar et al., (2015) carried out the experimental program as well as finite element models of three type of sandwich panel namely the corrugated, web, two directional corrugated core sandwich panel are analysed to get a better understanding of the behaviour of the three types of panels. It can be noted that the failure mode in CCP panel is due to intercellular buckling in the top skin and core at the line parallel to the support edge, in WCP panel the intercellular buckling phenomena occur at the top face under the point load, TCP panel, the intercellular buckling happened in the top skin and the core in diagonal direction.

Hai Fang et al., (2016) studied about a new form of composite sandwich panels, with steel plates as face sheets and bonded glass fiber-reinforced polymer (GFRP) pultruded hollow square tubes as core. the inference was that that the mid-span deflection and stress decreased, and the decent speed was getting smaller as the thickness of steel face sheet increases.

Gregory W. Kooistra et al., (2018) analysed the transverse compression and shear collapse mechanisms of a second order, hierarchical corrugated truss structure. The experimental investigation that has been conducted confirmed that analytical strength predictions of the second order truss, is found to be about ten times greater than that of a first order truss of the same relative density.

Amar Bayat et al., (2019) investigated the mechanical behaviour of composite sandwich insulated panels with AAC core material experimentally under quasistatic flexural three-point bending and indentation tests. A woven glass fibers plus aluminum sheets combination have been utilized as face skins reinforcement. Insulating AAC blocks with such composite sandwich sheets could, be a reliable option in reinforcing existing building components to improve its mechanical behaviour like housing safety as well as extension of their design life.

Hossein Malekinejad Bahabadi et al., (2020) manufactured three different geometries including simple foam core without any corrugated composite, longitudinal trapezoid, and square cores and investigated experimentally and numerically the resistance to separation between the skins of composite and different complex geometries of the corrugated core filled with PVC foam. The conclusion inferred is that the square core is concluded as an optimal design which has the highest bonding strength among other geometries.

Quanjin Ma et al.,(2021) The paper reviews recent trends in core structures and impact response of sandwich panels, classifying core types into foam-core, two- and three-dimensional periodic cores. It highlights advances in core designs, impact failure modes, and applications in aerospace, automotive, marine, and civil engineering. Future developments focus on optimizing mechanical performance, weight reduction, and impact resistance.

A. Hematian soorki et al., (2021) investigated the influence of reinforcement sheet on the bending strength of composite panel by numerical and experimental analysis based on three-point bending tests in both longitudinal and transverse directions. The results showed that for the longitudinal direction, adding a reinforcement sheet has an unfavourable effect on the bending strength to weight ratio of the panel, but for the transverse direction, it has been seen that it increases the bending strength to weight ratio significantly.

2.3 RESEARCH GAP

- Based on the above literatures, it has been found that the main disadvantage of the sandwich panel is low moment of inertia. So, we are performing a study to understand the correlation between varying flexural strength and varying core thickness and to improve core thickness.

- In this study, sandwich panels of varying core thickness has been studies.

2.4 SUMMARY

In this chapter, a summary of work carried out in published literature on sandwich structure is presented.

CHAPTER 3

MATERIALS AND METHODS

3.1 INTRODUCTION

A **composite material** is formed when two or more structurally distinct materials, which are generally insoluble in one another, are combined at the macroscopic level. These materials do not undergo a chemical reaction during the process, but rather, one material (the **matrix**) surrounds and reinforces another (the **reinforcement**). For example, in a typical composite, an **epoxy resin** may serve as the matrix, while materials like **glass fiber** or **carbon fiber** act as the reinforcement. The reinforcement fibers are the primary load carriers due to their high strength and stiffness, while the matrix, although weaker and lighter, helps to hold the fibers in place and transfers loads between them.

For a material to be classified as a **composite**, it must meet certain criteria. First, there must be **ease of availability** of the materials involved, making it practical for production and widespread use. Second, the composite must contain **two or more materials** that are distinct and visible at the macroscopic scale, ensuring that the individual components retain their identity and contribute to the overall material properties. The combination of these materials enhances the performance characteristics of the composite, such as strength, durability, and resistance to wear and fatigue, compared to the individual components alone.

3.2 MATERIALS

3.2.1 Glass Fibre:

Glass fiber is a material which is made from extremely fine fibers of glass. Glass fiber is a lightweight, extremely strong, and robust material. Although it is less stiff, the material is typically far less brittle, and the raw materials are less expensive when compared to other composite fibers. Glass Fiber Reinforced Polymer (GFRP) is a composite material which consists of glass fibers and a polymer matrix. The polymer matrix may be usually an epoxy or polyester thermosetting resin, or a vinyl ester. We have used glass fabric silane treated 10 mil cloth. Silane coupling agents are proved to be improving the mechanical strength of the composite materials, and for improving adhesion, and for improving resin modification and to improve surface modification.

3.2.2 PVC Foam:

PVC foam board is a leading member of the wide ranging and highly diverse family of polymers or plastics. It can be a solid or it can have an open cellular structure, in which case it is called foam, and the foams can be flexible or rigid. In simple explanation, the manufacturers make polyurethane foam by reacting polyols and diisocyanatos, both products derived from crude oil. A series of additives are considered necessary to produce high-quality PVC foam products, depending upon the required applications. We have used polyurethane foam of density 115kg/m3.

3.2.3 RESIN (Araldite GY 257)

Araldite GY 257 is a medium-viscosity, unmodified epoxy resin based on bisphenol-A. Possesses excellent mechanical properties and resistance to chemicals, which can be modified within its wide limits by using different types of hardeners as well as fillers. Araldite GY 257 is said to have a low tendency to crystallize. Used in aircraft and aerospace adhesives.

Properties of Epoxy resin GY 257:

- Resistance to crystallization
- Very good processing properties
- Good mechanical performances
- Good surface penetration
- Low viscosity: 500 - 650 MPa.s at 250 c
- Low Density: 1.15 g/cm3

3.2.4 HARDENER (Aradur C2963)

Aradur C2963 is a low viscosity, unfilled epoxy casting resin system, curing at room temperature. High filler addition possibility. It has key Properties like Good mechanical strength, and very good resistance to atmospheric and chemical degradation, Excellent electrical properties. Properties of hardener C2963:

Light yellow clear liquid with amine-like odour

- Density: 1000 Kg/m3
- Boiling point is more than 200 °C
- Flash point: 108 °C
- Kinematic viscosity: 30 – 70 MPa.S

Table 3.1 Materials and Preparation of Specimen

Construction	Material	Remark
Face sheet (Skin)	Glass fabric	Silane treated 10 mil cloth
Core	Divinycell PY Foam Board	Density – 115kg/m3
Resin	GY-257	Low viscosity and good penetration
Hardener	C-2963	Low viscosity

3.3 METHODOLOGY

3.3.1 Resin Infusion or Vacuum Infusion

Vacuum infusion (known as VARTM is a lamination process used in the manufacture of Fiber-Reinforced Plastic (FRP) parts. Dry materials are stacked onto a male or female mold surface and a thin plastic vacuum bag or semi rigid counter mold is sealed around the part perimeter. A vacuum pump is used to evacuate the air and to gradually apply atmospheric pressure to consolidate the dry materials and create a "vacuum cavity". Resin is then introduced into the system via strategically placed resin feeder lines. the pressure difference between the system and the outside atmospheric pressure pushes in the resin through the porous materials until the part is completely saturated. The vacuum is maintained until the part cures to ensure consolidation. Vacuum is maintained until the part hardens to ensure solidification. Infusion offers several advantages over traditional hand lay-up techniques. That protects that is Protecting workers and the environment from harmful volatiles Eliminates the confusion that can result from the release of organic compounds and the manual work of resins. It allows unlimited setup time because the resin will not get catalysed until all the materials get in place. The vacuum bag helps in promoting ideal resin content and distribution, as well as improved mechanical properties and provides low void content.

Figure 3.1 Vacuum Infusion Setup

3.3.2 Equipment and Material Requirements

Vacuum bagging requires some of the general materials to successfully pull vacuum and apply even pressure. They are:

Peel Ply - Against the floor of the remaining laminate, a peel ply or "RIP RAG" can be applied. Peel ply serves functions. First, it really works with getting rid of the want for sanding or grinding before secondary bonding operations.

Second, peel ply can be left at the laminate. This will assist to preserve it easy till the subsequent step of construction.

Bleeder Ply - Bleeder ply is taken into consideration as a light-weight blotter that is used to take in extra resins and core bonding putties, which "BLEEDS" via the peel ply.

Sealant Tapes - Sealant tapes are used to seal the bagging to the device or element.

Vacuum Valves - Vacuum valves are the attachment factors in among the bagging movie and the vacuum hose.

Vacuum Hose - -Vacuum hose can be bendy or rigid. Commonly, vacuum hose is manufactured from PVC tubing and is attached to the vacuum valve with the aid of using a quick run of bendy hose. Vacuum hose must be stiff sufficient such that it resists vacuum strain since it too could be evacuated.

Resin Traps - Resin traps are to be equipped withinside the vacuum hose among the element and the vacuum pump. This will assist in preventing any resin from migrating from the element to the pump.

Vacuum Pumps - Vacuum pumps are to be in several configurations and capacities. Most pumps used right here for evacuation device are electric powered pushed motor.

3.3.3 Infusion Procedure

With all the materials and equipment in place, the process can be started as Follows

1. Plug the end of cache infusion line with tacky tape, Vise grips, or another suitable clamp.

2. Turn on the vacuum pump. As air is evacuated Irom the bag, check for leaks and repair as necessary. A vacuum gage can be installed both near the pump and near the part. Catalyse the first resin source and any additional sources as needed.

The resin should be applied ahead of time to achieve the proper gel time. Usually, the resin will gel shortly after the part gets wet out.

3. Unplug the infusion line and insert into the resin source. The resin should easily be drawn through the line and begin to enter the part. Continue checking for leaks by listening and by watching for bubbles travelling through the laminate.

4. Continue Step 3 until all the lines have been used and the part is fully saturated. At that point ensure that every line have been plugged to ensure full vacuum

5. At this point it may be beneficial to reduce the pressure if possible. Excessive pressure may lead to squeezing out of too much resin and cause the laminate to be dry.

6. No vacuum bag can have a perfect seal, so leave the pump on until the resin has gelled or cured. In addition to drawing resin through the part, the vacuum bag helps to consolidate the part, ensuring that it gives better mechanical properties. 7. Once the part has cured, remove the vacuum bag, peel ply, bleeder, infusion lines, and dispose if necessary.

3.3.4 Advantages of Vacuum Infusion

- Vacuum bagging eliminates random bonding. Complete bonding of the core to facing will be structurally sound and hence will not produce blemishes due to voids.
- Vacuum bagging produces total attachment of core to facings which allows the sandwich to develop maximum mechanical properties.
- Vacuum bagging is easier on tools in that it produces very little stress on the tool itself. By contrast, stacking weights imposes considerable bending stresses on the tool.

3.3.5 SPECIMEN CUTTING: WATER JET MACHINING

A water jet cutter, also known as a water jet, is a tool which is capable of slicing into metal or composite materials using a jet of water at high velocity and pressure, or a mixture of water and an abrasive substance. The specified ASTM size 200mmx60mm for the bending testing is cut from the sandwich panels using the water jet machining process. It is the preferred method where the materials being cut are sensitive to the high temperatures.

CHAPTER 4

BENDING TEST

4.1 INTRODUCTION

The need to study the structural and failure characteristics of sandwich accoutrements and structures has surfaced during recent times because of increased acceptance of this structural conception in several critical weight Operations. Substantial quantum of structural sandwich mixes has been formerly used in numerous areas of aircraft, boat shells, wind turbine blades, coastal canvas platforms and ground balconies because of their superior structural capacity to carry transverse loads, superior bending stiffness, low weight, excellent thermal Sequestration and significant aural damping.

Out of several structural actions, the flexural property determination is basically needed in marine and aeronautical operations to determine the sandwich compound operation. Also, the demand of high stiffness and strength, substantially on flexural loads, together with low specific weight is an essential property on designing this recently developed compound. Still, the weakest point of similar compound rudiments consists in the possible debonding and delamination of the external guises of the sandwich skins, which must retain considerable Severity and strength, from the central part of the sandwich core. Also, it's needed to retain a low specific weight and acceptable shear stiffness. This chapter presents an experimental determination of flexural strength of sandwich panels of varying core thickness under three- point bending cargo which has been applied over the number of samples.

4.2 UNIVERSAL TESTING MACHINE(UTM)

A universal testing machine (UTM), also known as a universal tester, testing machine or accoutrements test frame, is used to test the tensile strength and compressive strength of accoutrements. An earlier name for a tensile testing machine is a tensiometer. The" universal" part of the name reflects that it can perform Numerous standard tensile and contraction tests on accoutrements, factors, and structures.

Figure 4.1 Universal Testing Machine

4.2.1 COMPONENTS

Load frame - Generally conforming of two strong supports for the machine. Some small machines have a single support.

Load cell - A force transducer or other means of measuring the load is needed. Periodic estimation is generally needed by governing regulations or quality system.

Cross head - A portable cross head (crosshead) is controlled to move over or down. Generally, this is at a constant speed occasionally called a constant rate of extension (CRE) machine. Some machines can program the crosshead speed or conduct cyclical testing, testing at constant force, testing at constant distortion, etc. Electromechanical, servo-hydraulic, direct drive, and resonance drive are used.

Means of measuring extension or deformation - -Numerous tests bear a measure of the response of the test instance to the movement of the cross head. Extensometers are occasionally used.

Output device - A means of furnishing the test result is demanded. Some aged machines have dial or digital displays and map reporters. Numerous newer machines have a computer interface for analysis and printing.

Conditioning - Numerous tests bear-controlled exertion (temperature, Moisture, pressure, etc.). The machine can be in a controlled room, or a special environmental chamber can be placed around the test instance for the test. Test fixtures, instance holding jaws, and related sample making outfit are called for in numerous test methods.

4.2.2 Application

The set-up and operation of a typical testing system for evaluating the mechanical properties of materials, especially in the context of **tensile testing**, are typically described in detail by standards organizations, such as ASTM or ISO. These standards provide guidelines for conducting tests, ensuring that the setup is consistent, accurate, and reproducible across various laboratories and testing environments. The system is designed to measure a material's response under load, particularly its **stretching (extension)** or **compression (contraction)** behavior, providing valuable insights into the material's **strength**, **ductility**, and **elasticity**.

In a typical testing system, the specimen, which is the material being tested, is placed between the **grips** of the testing machine. These grips hold the specimen securely while the test is being conducted. The specimen is typically elongated in a controlled manner to determine how it reacts to tensile stress. **Extensometers** are often used in such testing systems. These devices are designed to measure the **strain** of the specimen, that is, the change in length of the material as it is subjected to the applied load. Extensometers can automatically record the elongation (or contraction) of the specimen during the test. This helps to calculate key mechanical properties such as **Young's Modulus**, **ultimate tensile strength**, and **yield strength**, which are essential for understanding a material's performance under stress.

However, if an **extensometer** is not used, the testing machine itself can still measure the displacement between the **crossheads**—the parts of the machine that hold the specimen. The machine typically records the change in distance between these crossheads as the specimen undergoes elongation or contraction. This measurement is used to determine the amount of extension or contraction in the specimen, although it may be less precise than using an extensometer, especially for materials that experience very small or non-uniform deformations.

One of the key functions of the testing system is to ensure that all relevant factors affecting the specimen's behavior are recorded during the test. This includes not only the change in length of the specimen itself but also other critical parameters associated with the testing machine and its **drive systems**. These systems apply the load to the specimen, and any irregularities such as **slippage** of the specimen in the grips, or **non-uniform loading** due to imperfections in the machine, can be recorded. Such factors can influence the accuracy and reliability of the test results, and it is crucial for the system to capture them for proper analysis.

When the machine is started, it begins applying a steadily increasing load on the specimen. This loading process is usually controlled by a **feedback system** that ensures the test proceeds at a consistent rate, according to the test parameters defined in the testing standard. The machine's **control system** and associated **software** are responsible for recording both the **load** being applied to the specimen and the **extension** (or contraction) of the specimen as it reacts to the load. As the load is gradually increased, the machine continuously monitors and records the changes in both force and displacement.

Throughout the testing process, the **load-extension curve** is generated. This curve represents the relationship between the force applied to the specimen and the corresponding elongation or contraction. The data collected during the test can be used to generate a stress-strain curve, which provides a graphical representation of the material's **elastic behavior**, **yield point**, **ultimate strength**, and **fracture point**. The resulting data are crucial for evaluating the **mechanical properties** of the material, including its **ductility**, **tensile strength**, and **elastic modulus**. The machine's control software allows for the continuous monitoring of these parameters, and the data can be stored for later analysis or used in real-time to adjust test conditions if necessary.

In addition to measuring the primary parameters, such as load and elongation, the testing system can also monitor other variables, including temperature, strain rate, and environmental conditions. For instance, **temperature-controlled tests** are often performed to simulate real-world conditions where materials may be subjected to extreme temperatures. The software may also provide automated analysis, flagging anomalies or inconsistencies in the data, such as when the specimen slips in the grips or when unexpected behavior is observed.

After the test is complete, the machine typically provides a detailed output that includes all the data collected during the test. This output allows engineers and material scientists to analyze the material's performance and to determine whether it meets the necessary specifications for its intended application. Furthermore, the precise and automated nature of modern testing systems ensures that the results are reliable and reproducible, which is essential for quality control in manufacturing processes and material certification.

In summary, the operation of a tensile testing machine involves placing a specimen between grips, applying a steadily increasing load, and recording both the load and the extension (or contraction) of the specimen. The control system and associated software monitor all relevant factors, including any slippage in the grips, to ensure the accuracy of the test. Through this process, engineers can gather critical data on a material's mechanical properties, which are vital for designing products that will perform reliably under stress.

4.3 EXPERIMENTAL SETUP OF BENDING TEST

4.3.1 FACE SHEET SPECIMEN

Flexural testing was conducted following the **ASTM D 790-03** standard, titled "**Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials**." This standard provides a detailed procedure for determining the **flexural strength**, **flexural**

modulus, and the failure characteristics of both unreinforced and reinforced plastics, as well as electrical insulating materials. The test is designed to assess how materials perform when subjected to bending or flexural loads, which is essential for understanding their ability to withstand forces in applications such as structural components and load-bearing systems.

As per the **ASTM D 790-03** guidelines, the specimen used for the flexural test must have a **rectangular cross-section**. For consistency and accuracy in results, the final **dimensions** of the test specimens were standardized to a width of **25mm** and a length of **150mm**. These dimensions were chosen to fit the test equipment properly and to ensure that each specimen was subjected to uniform conditions. The rectangular shape of the specimen allows for accurate measurement of both the bending load and the corresponding deflection during the test.

During the testing process, a bending force was applied to the specimen, and the resulting **deflection** at the center of the beam was monitored. The maximum load the specimen could withstand before failure was recorded. From this data, key **flexural properties** such as **flexural strength**, **flexural modulus**, and **rigidity** were calculated. These properties are critical for evaluating how materials will perform under bending stresses in real-world applications. By adhering to the ASTM D 790-03 standard, the study ensured that the test results were reliable, reproducible, and could be compared effectively with other materials or configurations. This standardized approach is essential for ensuring the accuracy of the testing and the validity of the findings.

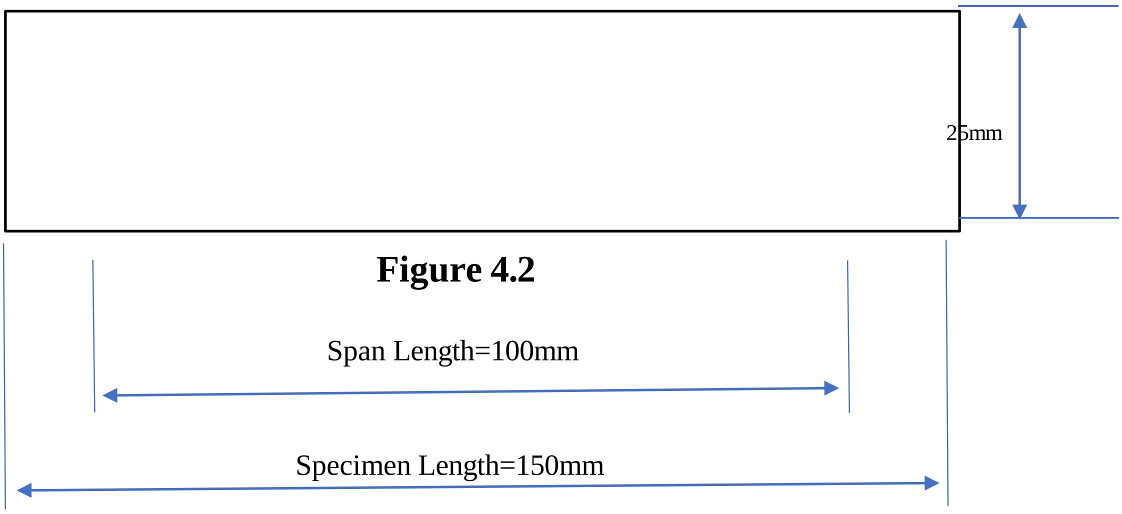


Figure 4.2 Dimensions of Face Sheet Specimen

The 3-point Bending test was conducted with a crosshead speed of 5mm/min using M100 (FSA) universal testing machine. The tests were performed with a test machine at a constant crosshead speed of 5mm/min.

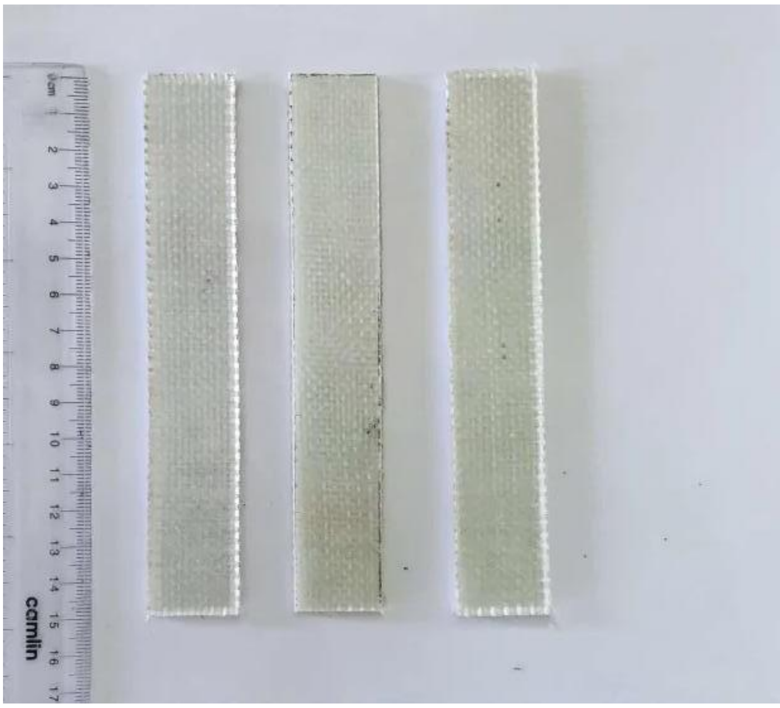


Figure 4.3 GFRP specimens before testing

4.3.2 SANDWICH PANEL SPECIMEN

We need to find bending strength of sandwich panels with core of 15mm(P1), 20mm(P2) and 25mm (P3)

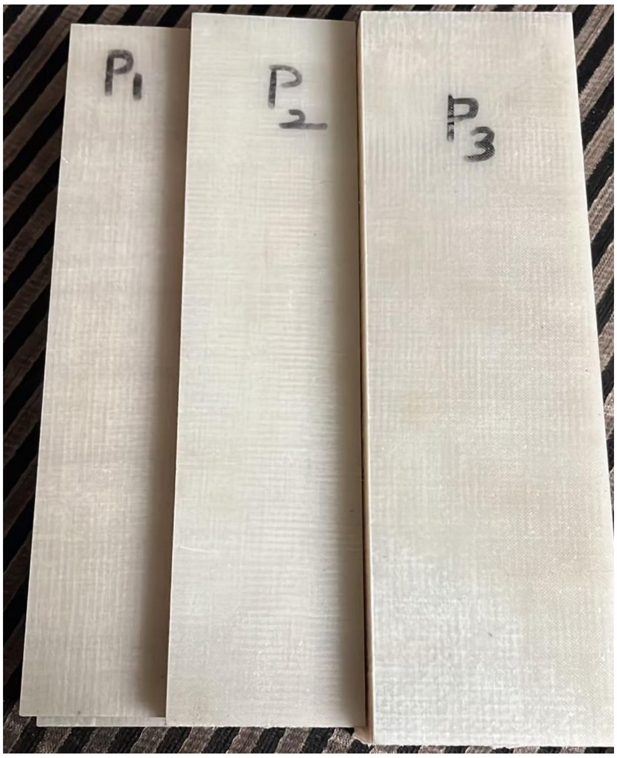


Figure 4.4 Sandwich panels with core thickness of 15mm, 20mm, 25mm respectively

Figure 4.5 Sandwich panels specimen before testing

CHAPTER 5
RESULT AND DISCUSSION

5.1 INTRODUCTION

This chapter presents the results of bending tests performed on face sheet and bending tests performed on sandwich panels of varying core thickness with core thickness of (15mm, 20mm, 25mm) performed in UTM(M-100) with a speed of 5mm/min.

5.2 EXPERIMENTAL RESULTS:

5.2.1 FACE SHEET

Flexural testing was carried out according to ASTM D 790-03 “Standard Test Methods for Flexural properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials”. As per the standard the test specimen cross section is rectangular in shape. The final dimensions were taken as 25mm wide and 150mm long.

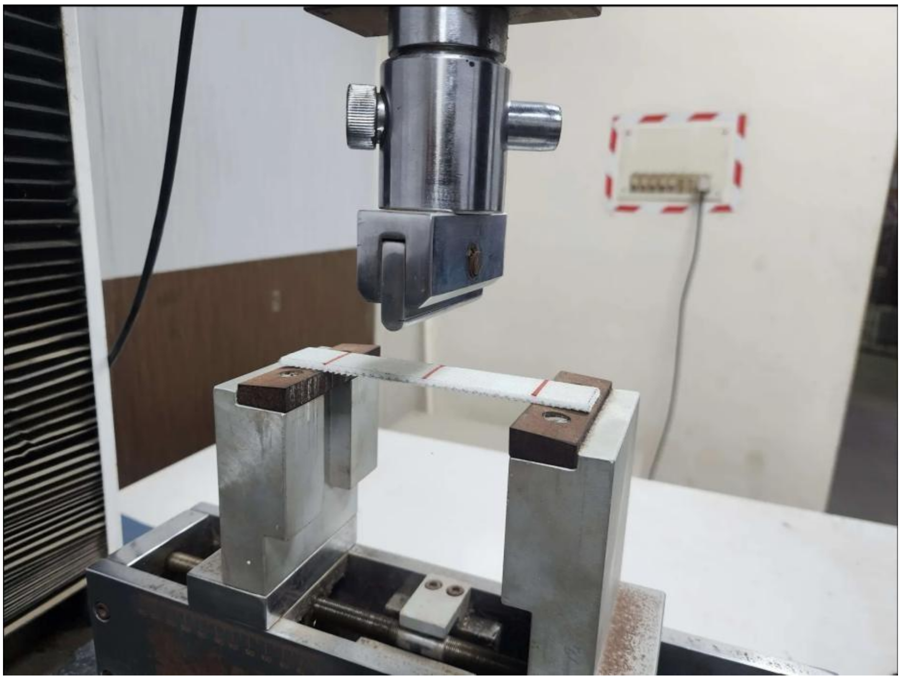


Figure 5.1 Flexural Test Setup for Face Sheet

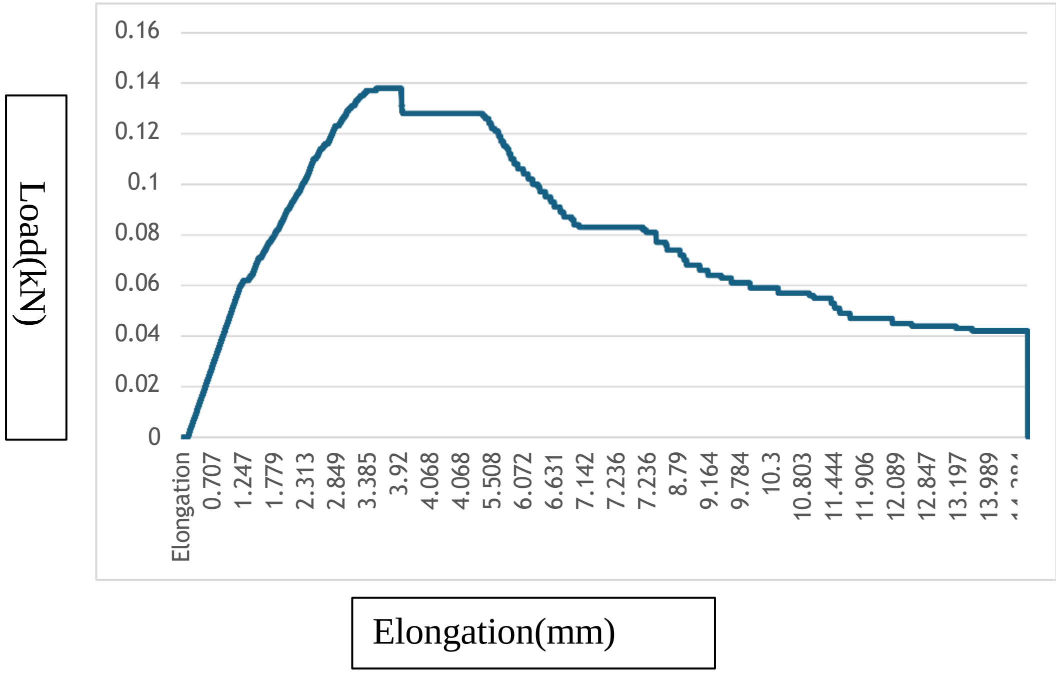


Figure 5.3: Load-Deflection Graph of flexural test of GFRP skin

Coupon Number	Width (mm)	Thickness (mm)	Elastic Modulus "Initial Slope "E (N/mm^2)	Mean (N/mm^2)	S.D	C.V	Bending Strength (N/mm^2)	Mean (N/mm^2)	S.D	C.V
B1	24.68	3.08	38.037				3772.91			
B3	24.68	3.13	33.21	35.578	1.90	0.0535	3377.03	3559.8	166.50	0.046
B5	24.72	3.2	36.167				3622.91			

Table 5.1 Flexural Strength of Face Sheet

Note: S.D = Standard Deviation C.V =Coefficient of Variation

The 3-point bending test was conducted with a crosshead speed of 5mm/min using universal testing machine as shown in Figure 5.1. The tests were performed with a test machine at a constant crosshead speed of 5 mm/min. Figure 5.3 shows the load displacement curves of the specimens. The bending strength was calculated by using the following equation

$$\sigma_f = 3PL/bd^2$$

σ_f = Bending stress (N/mm)

P = Maximum Load (N)

L = Span Length (mm)

b = Width of the specimen (mm)

d = Depth of the specimen (mm)

$$\delta = \frac{PL^3}{48EI}$$

δ = Deflection(mm)

P = Maximum Load (N)

L = Span Length (mm)

I = Moment of Inertia(mm^4)

5.2.2 SANDWICH PANEL

Flexural testing was carried out according to ASTM C 393-62 standard "Standard Test Method for Flexural Properties of Sandwich Constructions". This test determined the flexural strength of the sandwich panels. As per the standard the test specimen cross section is rectangular in length of the specimen is equal to the thickness of the sandwich shape. The construction, and the width shall be not less than twice the total thickness, nor half the span length. The specimen length is equal to the span plus 5 cm or one half the sandwich thickness whichever is greater.

The span length L_S, was calculated from Equation

$$L_s = \frac{2fF}{S}$$

L_S = Span Length(mm)

f = face sheet thickness

F = allowable facing stress

S = Core Shear stress

The **allowable facing stress (F)** of **193 MPa** for the glass fabric was determined through a **3-point bending test**, conducted in accordance with the **ASTM D790M** standard. This test is commonly used to evaluate the **flexural properties** of materials, including the **bending strength** and **modulus**. The **allowable core shear stress (S)**, which is **2 MPa**, was obtained from the **manufacturer's data sheet** for the specific core material used in the composite structure.

In the **3-point bending test**, the specimen is supported at two points, and a load is applied at the midpoint. The test measures how the material deforms under this applied load, which allows for the calculation of the bending strength and the modulus of the material. The **crosshead speed** during the test was set to **5 mm/min**, which ensures a controlled and steady application of load to the specimen. This speed is typically selected to allow for accurate measurements of the material's response to stress, ensuring the data collected is both reliable and reproducible.

The test was conducted using a **universal testing machine (UTM)**, which is designed to apply precise loads and measure the corresponding deformation of the specimen. A diagram of the setup can be seen in **Figure 5.4**. The results of the test provided critical data on the material's flexural behavior, which is essential for understanding its performance in real-world applications, such as aerospace, automotive, or structural engineering. These values, particularly the allowable stresses, are used to ensure the safety and performance of the composite material in its intended application

Figure 5.4 Experimental setup for sandwich panel

Figure 5.5 Flexural test on sandwich panel with core thickness of 20mm

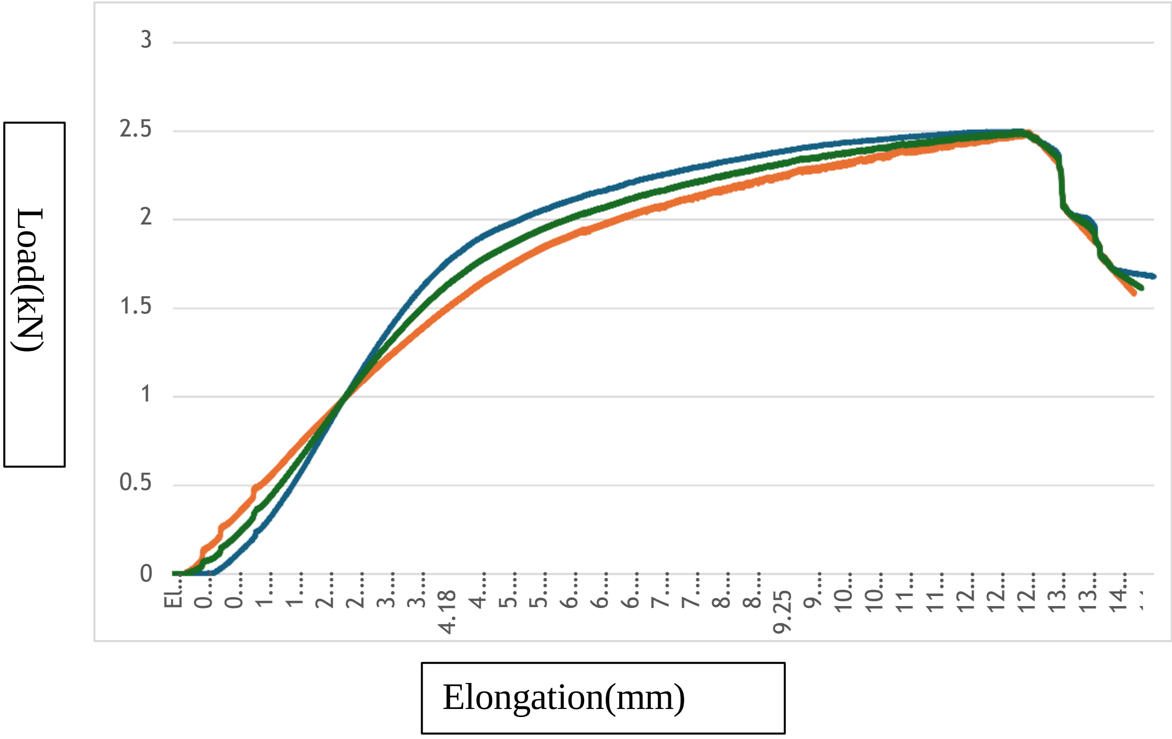


Figure 5.6 Graph of Load Vs Deflection of sandwich panel with core 15mm

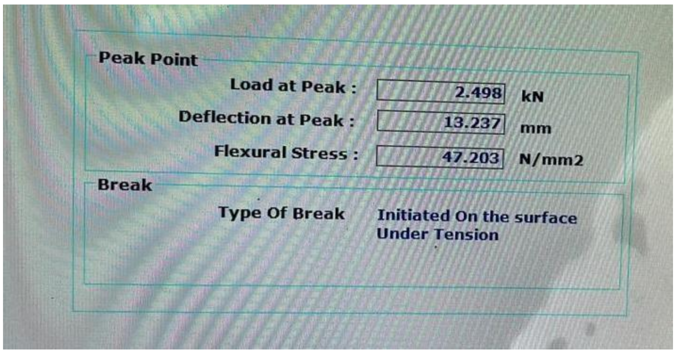


Figure 5.7 Experimental Values of 15mm core

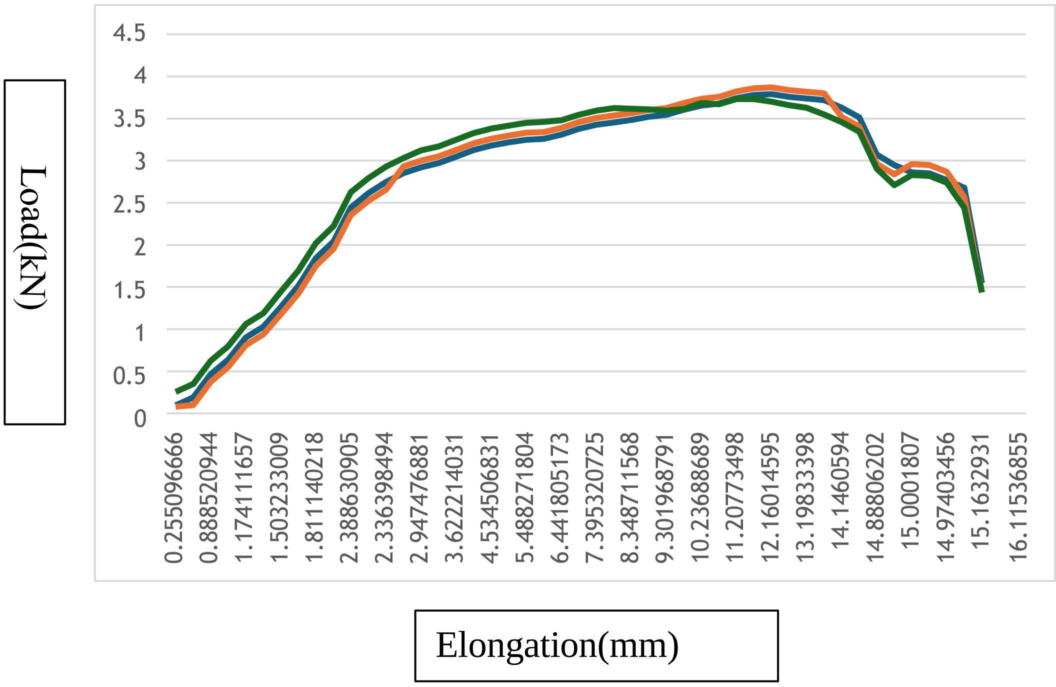
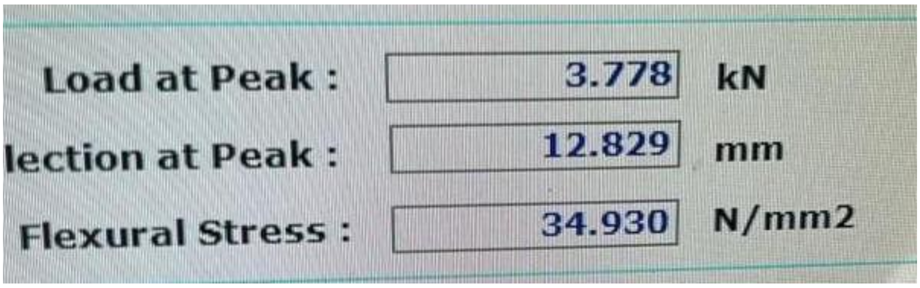


Figure 5.8 Graph of Load Vs Deflection of sandwich panel with core 20 mm



Load at Peak :	3.778	kN
lection at Peak :	12.829	mm
Flexural Stress :	34.930	N/mm2

Figure 5.9 Experimental Values of 20mm core

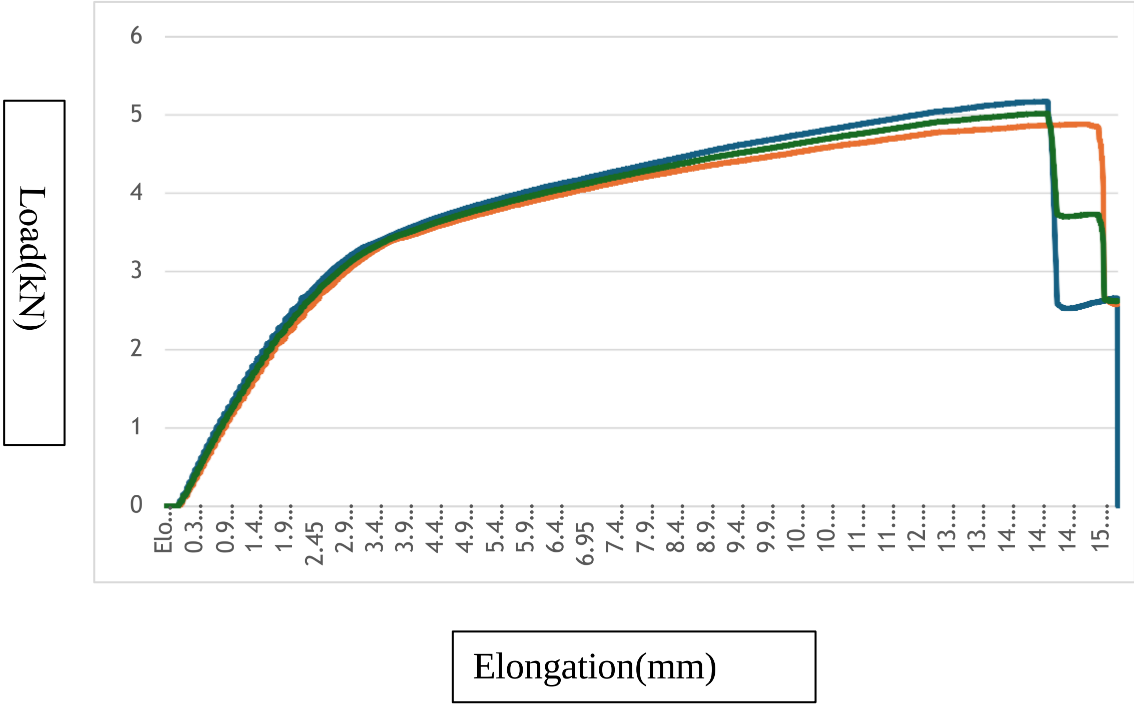
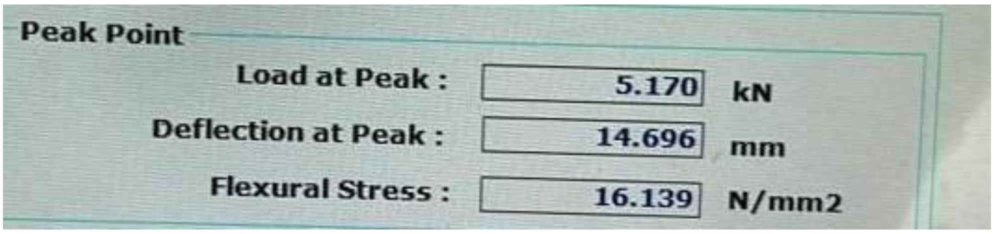


Figure 5.10 Graph of Load Vs Deflection of sandwich panel with core 25mm



Peak Point

Load at Peak :	5.170	kN
Deflection at Peak :	14.696	mm
Flexural Stress :	16.139	N/mm2

Figure 5.11 Experimental Values of 25mm core

Span Length (mm)	Width (mm)	Core thickness (mm)	Thickness (mm)	Load @ Peak (KN)	Deflection @ Peak (mm)	Flexural Stress (Mpa)	Flexural Rigidity (Nmm^2)	Weight (g)
150	45	15	21	2.498	13.237	47.203	34,728.75	115
150	60	20	26	3.778	13.829	34.930	87,880	180
150	75	25	31	5.170	14.696	16.139	186,193.75	217

Table 5.2 Flexural Strength of different core thickness sandwich panel

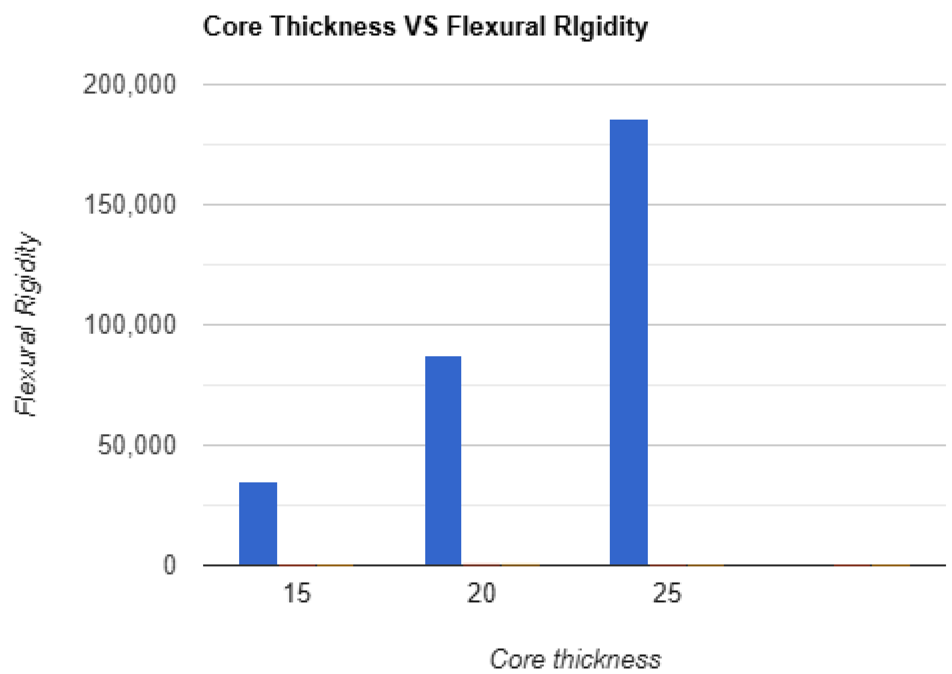


Figure 5.12 Bar graph of Core Thickness VS Flexural Rigidity

In this chapter the results of the experimental bending tests of glass fiber face sheet and flexural tests of sandwich panels of core thicknesses 15mm, 20mm and 25mm performed with a speed of 5mm/min are discussed.

Three different core thickness sandwich panel of core thickness varying from 15mm, 20mm and 25mm have shown a significant variance in flexural strength . It has been found that the specimen of core thickness of 15mm has a peak load of 2.498KN, the specimen of core thickness of 20mm has a peak load of 3.778KN and the 25mm core thickness specimen has a peak load of 5.17KN.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 SUMMARY

Composite materials, particularly Glass Fiber Reinforced Polymers (GFRP), are widely used in the aerospace industry due to their high strength-to-weight ratio, which is essential for minimizing weight while maintaining structural integrity. This study focuses on analyzing the flexural properties of GFRP panels and sandwich panels with varying core thicknesses (15mm, 20mm, and 25mm) to understand how the core thickness affects their flexural strength. The three-point bending test was performed on GFRP and sandwich panels, with the core material assumed to be a lightweight foam or honeycomb structure. The results revealed that the GFRP panel, a solid composite material, exhibited relatively high flexural strength due to the continuous glass fiber reinforcement. However, sandwich panels showed increased flexural strength with increasing core thickness. The 15mm core sandwich panels exhibited lower strength than the GFRP panel because the thinner core provided less bending resistance, while the 20mm core panels showed substantial improvement in strength due to the thicker core's ability to distribute loads more efficiently. The sandwich panels with a 25mm core provided the highest flexural strength, demonstrating the optimal performance of the thicker core in combination with the strong GFRP outer skins. A graph plotted from the experimental results showed a clear upward trend in flexural strength as the core thickness increased, with the 25mm core sandwich panels outperforming both the 15mm core panels and the solid GFRP panel. This study concludes that increasing the core thickness of sandwich panels significantly enhances their flexural strength, providing a better strength-to-weight ratio than solid GFRP panels, which is critical for aerospace applications. The findings suggest that thicker core sandwich panels offer a more efficient option for aerospace structures where weight reduction and strength are crucial. Further research could explore other core materials or configurations to optimize sandwich panel performance even more.

6.2 CONCLUSION

The Conclusion is as follows:

The **bending strength** of **Glass Fiber Reinforced Polymer (GFRP)** panels has been evaluated at different fiber orientations or angles, and the results have been analyzed and compared to understand how various orientations affect the material's performance under bending loads. This analysis is particularly important in composite materials like GFRP, where the orientation of fibers can significantly influence the mechanical properties, including **flexural strength** and **rigidity**. Additionally, the **flexural rigidity** of **sandwich panels** with varying core thicknesses—specifically 15mm, 20mm, and 25mm—was also investigated. The core material of these sandwich panels was assumed to be a lightweight and strong substance such as foam or honeycomb, which is commonly used in aerospace and other high-performance engineering applications.

The bending strength of sandwich panels with different core thicknesses was compared, and it was found that the specimen with a 25mm core thickness exhibited **higher flexural rigidity** than the other panels. Flexural rigidity is a measure of a material's resistance to bending under applied loads and is a critical factor in determining the overall stiffness of a structure. A higher flexural rigidity indicates a stiffer panel, which is desirable in many engineering applications where minimizing deflection under load is essential.

The study found that **sandwich panels with a 15mm core thickness** had a **flexural rigidity** of **34,728.75 Nmm^2**, while the **20mm core sandwich panel** showed a significantly higher rigidity value of **87,880 Nmm^2**. The **25mm core sandwich panel** exhibited the highest flexural rigidity of **186,193.75 Nmm^2**. These values indicate a substantial increase in the resistance to bending as the core thickness increases. Specifically, when the core thickness increased from 15mm to 20mm, the **flexural rigidity** increased by a factor of **2.53 times**. This means that the thicker core provides significantly more resistance to bending, contributing to the panel's overall stiffness.

Moreover, the study also revealed that the increase in flexural rigidity was not uniform across the core thicknesses. Between the **20mm core panel** and the **25mm core panel**, the **flexural rigidity** increased by **2.11 times**. This shows that while increasing the core thickness continues to improve the rigidity, the rate of increase becomes somewhat smaller between the 20mm and 25mm core thicknesses. Finally, when comparing the **15mm core panel** to the **25mm core panel**, the **rigidity increased by 5.36 times**, indicating that the greatest improvement in bending resistance occurs when moving from the thinnest core (15mm) to the thickest core (25mm).

These findings emphasize the importance of core thickness in the performance of sandwich panels, particularly in terms of **flexural rigidity**. The increased core thickness not only improves the bending strength but also contributes significantly to the overall **structural stiffness** of the panel. This is particularly relevant in aerospace applications, where components need to be as lightweight as possible without compromising their ability to withstand external loads and stresses. The ability to tailor the core thickness allows engineers to optimize the balance between weight and stiffness, ensuring the material performs effectively in its intended application.

From a design perspective, the results suggest that **sandwich panels with a 25mm core thickness** provide the most favorable combination of stiffness and strength, making them ideal for use in load-bearing applications that require high **flexural rigidity**. However, the trade-off between weight and performance must be carefully considered, as thicker cores may increase the overall weight of the panel, which could be a concern in certain applications, particularly in industries like aerospace where minimizing weight is a priority.

The comparison of flexural rigidity across different core thicknesses also highlights the potential for **sandwich panels** to outperform **solid materials** like GFRP in terms of **rigidity** and **strength-to-weight ratio**. By optimizing the core thickness, sandwich panels can be engineered to achieve specific performance goals while minimizing material use. This characteristic is particularly advantageous in industries such as aerospace, automotive, and construction, where the weight-to-strength ratio is a critical parameter.

In conclusion, this study demonstrated that **sandwich panels with thicker cores** exhibit significantly higher **flexural rigidity**, with the **rigidity increasing substantially** as the core thickness moves from 15mm to 25mm. The results underscore the importance of core thickness in determining the bending resistance and overall structural performance of sandwich panels. This information is valuable for engineers and designers who aim to optimize material properties for specific applications, particularly in high-performance industries such as aerospace, where both strength and lightweight construction are essential. Future research could explore the effects of different core materials and combinations of core and skin thicknesses to further enhance the performance characteristics of sandwich panels in a variety of structural applications.

INDIVIDUAL CONTRIBUTION

Magesh K: I took the lead in verifying the integrity of the data, ensuring that every piece was thoroughly examined for consistency and precision. By reviewing a wide range of academic articles and case studies, I built a strong framework for the project's success. I carefully cross-referenced various data sources to guarantee its accuracy, which played a key role in the project's development. My extensive research provided the necessary insights to shape the project's direction and ensure its reliability.

Srivyas Narayan: I took charge of identifying and procuring the necessary materials to support the project's goals. This involved collaborating with multiple vendors to ensure timely delivery and quality assurance. I also maintained a detailed inventory to track all materials, ensuring no shortages or delays. My role required thorough research to find the most cost-effective yet reliable sources for the materials needed. Through careful planning and negotiation, I was able to secure the resources necessary for the successful completion of the project.

Muthiah R: I compiled the data and generated detailed report that accurately reflected the findings of the experiment. I ensured that every step of the experiment adhered strictly to the provided by our guide and ASTM standards. This involved cross-referencing each procedure to confirm compliance with industry best practices. I also worked closely with the team to address any discrepancies and make necessary adjustments during the process. My attention to detail in both reporting and maintaining standards contributed to the reliability and validity of the experiment's results.

References

1. William S. Slaughter, Jianqiang Fan and Norman A. Fleck1 T “Dynamic Compressive Failure of Fiber Composites” International Journal of Solids and Structures 18 March 1996

2. F. Coˆte, V.S. Deshpande, N.A. Fleck, A.G. Evans “The compressive and shear responses of corrugated and diamond lattice materials” International Journal of Solids and Structures 31May 2005

3. benjaminp. Russell, Adammalcom, Haydnn. G. Wadleyandvikrams. Deshande “Small perforations in corrugated sandwich panel significantly enhance low frequency sound absorption and transmission loss” International Journal of Solids and Structures 6 September 2017

4. Benjaminp. Russell, Adammalcom, Haydnn. G. Wadleyandvikrams. Deshande “Dynamic Compressive Response of Composite Corrugated Cores” Journal of Mechanics of Materials and Structures Vol. 5, No. 3, 2010

5.

6. V. Rubino, V.S. Deshpande n, N.A. Fleck “The three-point bending of Y-frame and corrugated core sandwich beams” Journal of Mechanics of Materials and Structures Vol .2010

7. S. Kazemahvazi and D. Zenkert ICSS 8 A. J. M. Ferreira (Editor) © FEUP, Porto, “The Compressive and Shear Responses of Corrugated Hierarchical and Foam Filled Sandwich Structures” 8th International Conference on Sandwich Structures 2008

8. L. Valdevit , J.W. Hutchinson , A.G. Evans “Structurally optimized sandwich panels with prismatic cores” International Journal of Solids and Structures 41 (2004) 5105–5124

9. M. Nawar1, B. Eltaly2 and K. Kandeel “Structural Performance of Sandwich Panel Under Point Load” Engineering Research Journal Faculty of Engineering Minoufiya University Engineering Research Journal, Vol. 38, 57 No.1, January 2015, PP: 57-64. © Faculty of Engineering, Minoufiya University, Egypt.

9.A. Vaziri, Z. Xue and J. W. Hutchinson “Metal Sandwich Plates with Polymer Foam-Filled Cores” Journal of Mechanics of Materials and Structures Vol. 1, No. 1, 2006

10. A. Hemmatian Soorki, A. Kabiri Ataabadi, M. Loh-Mousavi "Experimental and Numerical Investigation of Reinforcement Sheet Effect on the Bending Strength of Corrugated Composite Panels" Mechanics of Advanced Composite Structures

11. Hossein Malekinejad Bahabadi, Amin Farrokhabadi, Gholam Hossein Rahimi "Investigation of debonding growth between composite skins and corrugated foam-composite core in sandwich panels under bending loading"

12. Fukun Xia, Yvonne Durandet, T X Yu2 and Dong "Large deformation of corrugated sandwich panels under three-point bending" Rua Journal of Sandwich Structures and Materials

13. Denis D. Cartie´, Norman A. Fleck "The effect of pin reinforcement upon the through-thickness compressive strength of foam-cored sandwich panels" 2003

14. Gregory W. Kooistra Vikram Deshpande UK Haydn N. G. Wadley "Hierarchical Corrugated Core Sandwich Panel Concept" Journal of Applied Mechanics March 2007 Vol 74/259

15. Shujuan Hou, Shuyun Zhao, Lili Ren, Xu Han, Qing Li "Crashworthiness optimization of corrugated sandwich panels" Journal of Material and Design (2013) 1071-1084.

16. Iman Dayyani & Saeed Ziaei-Rad & Hamid Salehi "Numerical and Experimental Investigation on Mechanical Behaviour of Composite Corrugated Core Received: 8 June 2011 / Accepted: 24 November 2011 / Published online: 20 December 2011 Springer Science Business Media B.V. 2011

17. Shujuan Houa, Chengfu Shua, Shuyun Zhaoa, Xu Hana, Qing Lic "Experimental and Numerical Studies on Multi-Layered Corrugated Sandwich Panels Under Crushing Loading" Journal of Composite Structures (2006)

18. Jian Xiong, Li Ma, Ariel Stocchi, Jinshui Yang, Linzhi Wu, Shidong Pan "Bending response of carbon fiber composite sandwich beams with threedimensional honeycomb cores" Journal of Composite Structure (2013)

19. Dan Zenkert, Magnus Burman "Failure Mode Shifts in Fatigue of Sandwich Beams" International Conference on Composite Materials January 2009

20. Y.S. Tian, T.J. Lu "Optimal Design of Compression Corrugated Panels" ThinWalled Structures 43 (2005) 477-498.

21. Jin Zhang, Peter Supernak, Simon Mueller-Alander, Chun H. Wang "Improving the Bending Strength and Energy Absorption of Corrugated Sandwich Composite Structures" Materials and Design 52 (2013) 767-773

22. Nathan Wicks, John W. Hutchinson "Performance of sandwich plates with truss cores" Mechanics of Materials 36 (2004) 739–751

23. Amar Bayat, Gholamhossein Liaghat, Hadi Sabouri, Ghasem Dehghani Ashkezari, Ehsan Pedram, Sayed Ahmad Taghizadeh, Muhammad Kashif Khan and Omid Razmkhah "Experimental investigation on the quasi-static mechanical behavior of autoclaved aerated concrete insulated sandwich panels" Journal of Sandwich Structures and Materials 0(0) 1–30! The Author(s) 2019

24. Hai Fang, Huiyuan Shi, Yue Wang, Yujun Qi, and Weiqing Liu"Experimental and Theoretical Study of Sandwich Panels with Steel Face sheets and GFRP Core" Advances in Materials Science and Engineering Volume 2016, Article ID 7159205.

25. Jaafar Mohammed Ali Moheb, Ehsan Sabah Al-Ameen "Study of the Bending Characteristics in Composite Sandwich Structures" IOP Conference Series: Materials Science and Engineering 2020.